



FCL ET AL WORKMAN HZ

3B13-20/4B10-19-1-31W1M

Well Data Report

July 30, 2011

Prepared for:

Federated Co-operatives Limited

c/o Adonai Resources Corporation

Suite 300, 2631, 28th Ave.

Regina, Sask, Canada

S4S 6X3

License: 11C657

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WELL DATA

WELL NAME: FCL et al Workman Hz
 LOCATION: 3B13-20/4B10-19-1-31W1M
 CO-ORDINATES: SURFACE: 252.0 m S of N, 71.0 m E of W boundaries, Sec 20
 BOTTOM: 280.70 m S, 493.13 m W of Well Centre
 OPERATOR: **Federated Co-operatives Limited**
 PARTICIPANTS: **Adonai Resources Corporation**
 DRILLING LICENCE: 11C657
 PARTICIPANTS: N/A
 ELEVATIONS: GROUND: 500.07 m K.B.: 505.5 m
 TOTAL DEPTH: 1650.0 m MD/1188.0 m TVD/-682.4 m SS
 SPUD DATE: 17:45 hrs, 24/07/11
 DRILL OUT: Surface Casing 01:00 hrs, 26/07/11
 Intermediate Casing N/A
 FINISHED DRILLING: 0:10 hrs, 30/07/11
 SECURITY STATUS: Confidential FIELD: Workman
 OBJECTIVE: Sherwood Porosity
 PROVINCE: Saskatchewan AREA: Southeast
 CASING: Surface: 15 jts 244.5 mm, 48.1 kg/m H-40 csg.;
 landed @ 200.0 m KB
 Intermediate: N/A
 COMPLETION: Horizontal development well
 DRILL CONTRACTOR: Advance drilling Ltd. RIG # 8 TOOLPUSH: Garth Brisbois
 WELL ENGINEER: Eddie Hackman

WELL GEOLOGIST: Jeff Bovee/Dean Polard

SERVICE CO'S:	DRILLING MUD:	Prairie Mud Service
	DIRECTIONAL:	Phoenix Technology services
	CEMENTING:	BJ Services
	HOLE SURVEYS:	N/A
	LOGGING:	N/A

GEOLOGICAL MARKERS**KB = 505.5 m**

FORMATION TOPS	PROGNOSIS (M SS)	SAMPLE DEPTH			DIFF.
		M DEPTH (M)	TV DEPTH (M)	ELEVATION (M SS)	
<i>Jurassic</i>					
Upper Watrous	-590.0	-	-	-	-
Lower Watrous Red Beds	-622.5	1147.0	1127.0	-621.5	1.0
<i>Mississippian</i>					
Frobisher Evaporite	-658.5	1207.4	1160.0	-654.5	4.0
Frobisher State A.	-661.5	1227.0	1166.5	-661.0	0.5
Griffin (Hastings Evaporite)	-663.5	1236.5	1169.0	-663.6	-0.1
Griffin Anhydrite	-669.5	1281.5	1177.5	-672.0	-2.5
<i>Sherwood argillaceous Marker</i>	-673.5	1320.0	1181.9	-676.4	-2.9
<i>Landing Point</i>	-677.5	1327.0	1182.7	-677.2	0.3
<i>Sherwood porosity</i>	-677.5	1378.0	1187.0	-681.5	-4.0
<i>TD Leg #1</i>	-686.0	1650.0	1188.0	-682.4	3.6

N.B. True vertical and subsea depths are based on Phoenix directional surveys.

* Tops based on ROPs

MUD RECORD

Mud Type KCL

Mud Company Prairie Mud Service

Mudman Ian Scott

Hole Diameter Build Section: 222 mm
Lateral Section: 200 mm

Casing O.D./I.D. Surface: 244.5 mm/228.6 mm
Intermediate: N/A

Casing set at Surface: 200.0 m KB
Intermediate: N/A

Pump Liner Size #1 140 mm Stroke Length 229 mm
#2 Stroke Length

Volume Per Pump Stroke #1 0.0104 m³
#2

Pump Output #1 1.3 m³/min @ 130 Stk/min (97.5 % efficiency) (Build & Lateral)

Sample Interval 5 m samples washed and vialled from 1140 to 1650 m MD;

Sample Quality poor through build & lateral sections

Mud Gas: Mud gas logged from ~1075 m MD to TD of well with a Pason total gas detection system.

Date	Depth (m)	Density (Kg/m ³)	Viscosity (sec/l)	Fluid Loss (cm ³)	Filter Cake (mm)	Ph	Chlorides (mg/L)
July 27/11	1103	1075	30	11.2	<1	9.0	2000
July 28/11	1103	1075	36	9.2	<1	9.0	18000
July 29/11	1410	1085	33	8.2	0.1	7.5	18000

BIT RECORD

Bit	Size	Type	Jets	Out	Meters	Hours	ROP (m/hr)
1A	349	GF05	-	200	200	8.5	-
2	222	QD405F	7/14	1327	1127	48	-
1Hz	200	QD404F	7/11.1	1650	323	14.9	-

STRAPS

Strap No.	Strap (m)	Pipe Tally (m)	Difference (m)
1(ICP)	-	-	-
2	-	-	-

SURVEYS (PRE-DIRECTIONAL)***SURFACE HOLE**

Depth (m)	Deviation (°)
40	0.36
75	0.31
110	0.63
145	1.05

MAIN HOLE

Depth (m)	Deviation (°)	Azimuth (°)
<i>See Matrix Surveys</i>		

* As read by drillers.

MATRIX DIRECTIONAL SURVEYS**BUILD AND LATERAL SECTIONS**

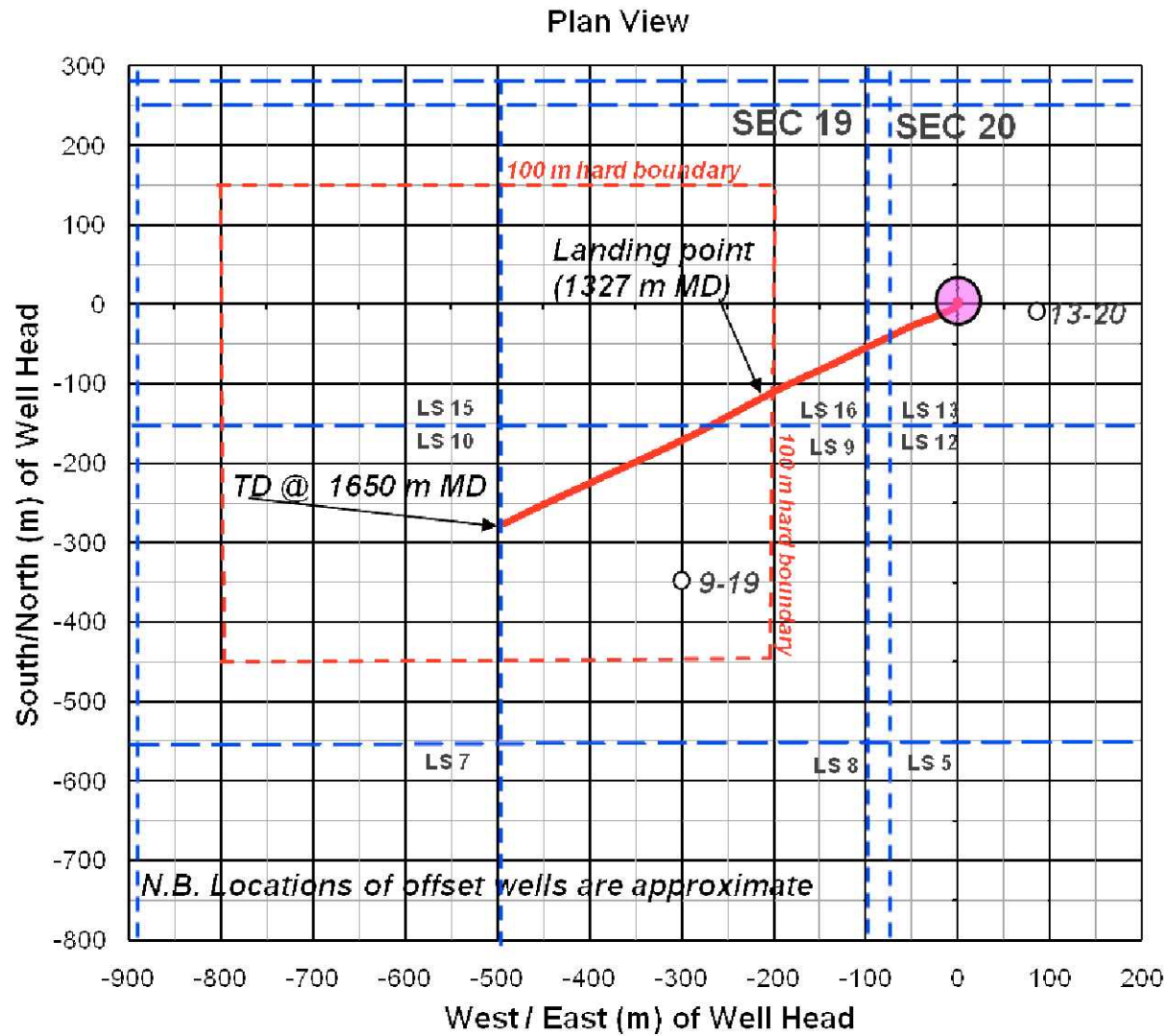
If Co-ordinates are Negative, Direction is S, W.

Measured Depth Meters	Incl Angle Deg	Drift Direction Deg	TRUE Vertical Depth	Subsea TVD Meters	N-S Meters	E-W Meters	Vertical Section Meters	BUILD RATE Deg/30	Dogleg Severity Deg/30
200	0	0	200	305.53	0	0	0	0	0
226.74	1.1	228	226.74	278.79	-0.17	-0.19	0.25	1.23	1.23
282.45	1.9	247.5	282.43	223.1	-0.88	-1.44	1.69	0.51	0.43
338	0.6	23.3	337.97	167.56	-0.97	-2.18	2.37	1.28	-0.7
393.55	1.8	283	393.51	112.02	-0.5	-2.91	2.79	1.08	0.65
449.08	0.3	219.1	449.03	56.5	-0.42	-3.85	3.57	0.91	-0.81
504.53	0.8	116.8	504.48	1.05	-0.71	-3.6	3.49	0.49	0.27
560.25	1.2	106.6	560.19	-54.66	-1.05	-2.69	2.86	0.23	0.22
616.29	1.7	114.4	616.21	-110.68	-1.56	-1.37	1.96	0.29	0.27
673.57	0.6	263.1	673.49	-167.96	-1.95	-0.9	1.73	1.17	-0.58
730.67	1.3	195.3	730.58	-225.05	-2.61	-1.37	2.46	0.63	0.37
787.89	1.5	174.6	787.78	-282.25	-3.98	-1.47	3.22	0.28	0.1
845.06	1.7	234.3	844.93	-339.4	-5.22	-2.08	4.37	0.84	0.1
902.11	0.6	144.7	901.97	-396.44	-5.96	-2.6	5.17	0.95	-0.58
940	0.6	56.9	939.86	-434.33	-6.01	-2.32	4.96	0.66	0
949.54	0.5	69.5	949.4	-443.87	-5.97	-2.24	4.86	0.49	-0.31
959.09	1.1	215.8	958.95	-453.42	-6.03	-2.25	4.91	4.84	1.88
KOP @ 956 m									
968.62	3.6	224.4	968.47	-462.94	-6.32	-2.52	5.28	7.93	7.87
978.07	6.2	228.9	977.89	-472.36	-6.87	-3.11	6.06	8.34	8.25
987.6	8.6	232.2	987.34	-481.81	-7.64	-4.06	7.27	7.67	7.56
997.11	10.9	236.5	996.71	-491.18	-8.57	-5.37	8.87	7.61	7.26
1006.67	12.4	240.7	1006.07	-500.54	-9.57	-7.02	10.8	5.4	4.71
1016.24	14.5	242.8	1015.38	-509.85	-10.63	-8.98	13.02	6.76	6.58
1025.8	16.7	243.3	1024.59	-519.06	-11.79	-11.27	15.59	6.92	6.9
1035.35	18.9	242.2	1033.68	-528.15	-13.13	-13.87	18.51	6.99	6.91
1044.81	21.1	241.5	1042.57	-537.04	-14.65	-16.72	21.74	7.02	6.98

1054.27	23.4	241.7	1051.32	-545.79	-16.36	-19.87	25.32	7.3	7.29
1063.97	25.8	242.5	1060.14	-554.61	-18.25	-23.44	29.36	7.49	7.42
1073.52	28.3	245.2	1068.65	-563.12	-20.16	-27.34	33.7	8.75	7.85
1083.1	30.8	246	1076.98	-571.45	-22.11	-31.64	38.4	7.93	7.83
1092.7	32.4	247.9	1085.16	-579.63	-24.07	-36.27	43.4	5.89	5
1102.23	34.9	246.7	1093.09	-587.56	-26.11	-41.14	48.65	8.14	7.87
1111.68	37.2	244.3	1100.73	-595.2	-28.42	-46.2	54.19	8.57	7.3
1121.16	39.5	242.8	1108.16	-602.63	-31.04	-51.46	60.07	7.85	7.28
1130.67	42	241	1115.37	-609.84	-33.97	-56.94	66.27	8.71	7.89
1140.22	44.6	240	1122.32	-616.79	-37.19	-62.64	72.82	8.45	8.17
1149.67	47.4	240	1128.88	-623.35	-40.59	-68.52	79.62	8.89	8.89
1159.14	50.5	240.3	1135.1	-629.57	-44.15	-74.72	86.76	9.85	9.82
1168.69	53.6	241.6	1140.97	-635.44	-47.8	-81.3	94.29	10.26	9.74
1178.16	57	241.7	1146.36	-640.83	-51.5	-88.15	102.07	10.77	10.77
1187.61	59.9	241.3	1151.31	-645.78	-55.34	-95.23	110.12	9.27	9.21
1197.15	63.3	240.6	1155.84	-650.31	-59.42	-102.56	118.51	10.87	10.69
1206.72	67.1	240.2	1159.86	-654.33	-63.71	-110.11	127.2	11.97	11.91
1216.26	70.7	239.6	1163.29	-657.76	-68.17	-117.81	136.1	11.46	11.32
1225.77	73.3	239.6	1166.23	-660.7	-72.75	-125.61	145.14	8.2	8.2
1235.31	75	240.8	1168.83	-663.3	-77.31	-133.58	154.31	6.46	5.35
1244.87	76.8	242	1171.16	-665.63	-81.74	-141.72	163.59	6.73	5.65
1254.32	78.8	242.4	1173.16	-667.63	-86.05	-149.89	172.82	6.47	6.35
1263.9	80.4	242.4	1174.89	-669.36	-90.42	-158.24	182.24	5.01	5.01
1273.47	81.4	242.1	1176.4	-670.87	-94.82	-166.6	191.68	3.27	3.13
1282.94	82.7	240.9	1177.71	-672.18	-99.29	-174.84	201.06	5.58	4.12
1292.5	83.2	240.7	1178.89	-673.36	-103.92	-183.12	210.55	1.69	1.57
1302.04	83.6	240.2	1179.98	-674.45	-108.59	-191.37	220.03	2.01	1.26
1311	83.7	240.1	1180.97	-675.44	-113.03	-199.09	228.93	0.47	0.33
1320.36	83.8	238.4	1181.99	-676.46	-117.78	-207.09	238.23	5.43	0.32
Landing point @ 1327 m									
1329.88	83.9	238.2	1183.01	-677.48	-122.76	-215.14	247.69	0.7	0.32
1339.41	84.1	238	1184.01	-678.48	-127.77	-223.19	257.15	0.89	0.63
1348.93	85.2	238.1	1184.9	-679.37	-132.78	-231.23	266.62	3.48	3.47
1358.49	85.9	238	1185.64	-680.11	-137.83	-239.32	276.14	2.22	2.2
1368.05	86.1	237.7	1186.31	-680.78	-142.9	-247.39	285.67	1.13	0.63
1377.61	85.6	237.3	1187	-681.47	-148.02	-255.43	295.18	2.01	-1.57

1387.16	85.1	237	1187.77	-682.24	-153.19	-263.43	304.68	1.83	-1.57
1396.62	86	237.7	1188.51	-682.98	-158.27	-271.37	314.1	3.61	2.85
1406.07	87.9	239.3	1189.01	-683.48	-163.2	-279.41	323.53	7.88	6.03
1415.62	89.9	240.6	1189.19	-683.66	-167.99	-287.68	333.07	7.49	6.28
1425.16	90.4	240.8	1189.17	-683.64	-172.65	-296	342.61	1.69	1.57
1434.7	90.6	240.2	1189.08	-683.55	-177.35	-304.3	352.15	1.99	0.63
1444.05	90.5	240.3	1188.99	-683.46	-181.99	-312.42	361.5	0.45	-0.32
1453.6	90.2	241.2	1188.94	-683.41	-186.66	-320.75	371.05	2.98	-0.94
1463.15	90.2	242	1188.9	-683.37	-191.2	-329.15	380.6	2.51	0
1472.72	90.2	241.9	1188.87	-683.34	-195.7	-337.6	390.17	0.31	0
1482.19	90.4	241.7	1188.82	-683.29	-200.17	-345.94	399.64	0.9	0.63
1491.76	91.7	242	1188.64	-683.11	-204.69	-354.38	409.2	4.18	4.08
1501.32	92.3	242.3	1188.31	-682.78	-209.15	-362.83	418.75	2.1	1.88
1510.78	91.8	242.3	1187.97	-682.44	-213.55	-371.2	428.2	1.59	-1.59
1520.35	90.8	241.6	1187.75	-682.22	-218.05	-379.64	437.77	3.83	-3.13
1529.82	89.4	241.4	1187.74	-682.21	-222.56	-387.96	447.24	4.48	-4.44
1539.29	89.5	241.3	1187.83	-682.3	-227.1	-396.27	456.71	0.45	0.32
1548.76	90	241.7	1187.87	-682.34	-231.62	-404.6	466.18	2.03	1.58
1558.23	90.7	242.1	1187.81	-682.28	-236.08	-412.95	475.65	2.55	2.22
1567.78	90.4	241.6	1187.72	-682.19	-240.59	-421.37	485.19	1.83	-0.94
1577.34	90.5	241.6	1187.65	-682.12	-245.14	-429.78	494.75	0.31	0.31
1586.81	90.7	241.6	1187.55	-682.02	-249.64	-438.11	504.22	0.63	0.63
1596.38	90.1	241.5	1187.48	-681.95	-254.2	-446.52	513.79	1.91	-1.88
1605.94	89.8	241.1	1187.49	-681.96	-258.79	-454.91	523.35	1.57	-0.94
1615.51	89.2	240.7	1187.57	-682.04	-263.44	-463.27	532.92	2.26	-1.88
1624.96	89.5	240.2	1187.68	-682.15	-268.1	-471.49	542.37	1.85	0.95
1634	89.4	239.8	1187.77	-682.24	-272.62	-479.32	551.41	1.37	-0.33
TD @ 1560 m									
Extrap. To bit									
1650	89.3	239.6	1187.95	-682.42	-280.7	-493.13	567.4	0.42	-0.19

FCL et al Workman Hz 3B13-20/4B10-19-1-31W1M



RESERVOIR QUALITY SUMMARY¹

Reservoir Quality	From (m MD)	To (m MD)	Length (m)	Total (m)
Fair (F)	1370	1395	25	105
	1490	1570	80	
Poor-Fair (P-F)	1460	1490	30	110
	1570	1650	80	
Poor (P)	1360	1370	10	75
	1395	1460	65	
Very Poor (VP)	1345	1360	15	15
Non –Very Poor (N - VP)	1327	1345	20	20
<i>Total open hole below 1325.0 m MD</i>	1325	1650	325	325

Interval (m MD)	Total Gas (%)	ROP (min/m)	Oil Stain/ Flor (%)	Reservoir Quality	Lithology, Markers
1325-1345	1-3	2-5	NIL	N - VP	Dolomite
1345-1360	2-4	2.5	TRC	VP	Limestone/Dolomite
1360-1370	2-4	2.5	TRC	P	Limestone/Dolomite
1370-1395	3-40	1.2-2	5-30	F	Limestone/Dolomite
1395-1460	3-12	2-2.5	5-10	P	Limestone
1460-1490	4-6	2-2.5	10	P-F	Limestone/Dolomite
1490-1570	10-65	2	20	F	Limestone/Dolomite
1570-1650	5-15	2	10-20	P-F	Limestone/Dolomite

¹ Assessment of reservoir quality is based on several objective and subjective parameters. Objective parameters include ROP and Gas readings. Subjective parameters include visual estimations of porosity, permeability and oil staining. These estimations are, in turn, affected by sample quality, including actual sampling technique, sample lag times and the common presence of backhole cavings. As such, a summary of reservoir quality is for guidance purposes only.

GEOLOGICAL SUMMARY

This horizontal ‘monobore’ well was drilled to develop the Frobisher (Sherwood) Beds in the Workman area of southeastern Saskatchewan. The well was ‘kicked off’ from the vertical at 956 m KB (19:15 hrs, July 26/11). The build section was drilled with a 2.3 degree bent mud motor assembly and PDC bit. Sample cuttings were collected at 5 m intervals from 1140 m MD to TD. Sample quality was poor to fair throughout the build and lateral section of the well.

The top of the Mississippian erosional surface (Frobisher Evap.) was encountered at c. 1207.4 m MD (1160.0 m TVD / -654.6 m SS), about 4.0 m above the prognosed depth. The intended target landing point was immediately above the Sherwood porosity. Due to the depth of the Mississippian erosional surface the target was presumed to also be higher than the prognosed depth, and was subsequently raised by 1.5 m. The Frobisher (State A) was encountered at c. 1227.0 m MD (1166.5 m TVD / -661.0 m SS) 0.5 metres *above* the prognosed depth. The Griffin (Anhydrite) was encountered at c. 1281.5 m MD (1177.5 m TVD / -672.0 m SS) 2.5 m *below* the prognosed depth; thus revealing a thickening of the Frobisher at this location. Subsequently the target was lowered, and the remainder of the landing was rotated at 82 degrees (building angle under rotation to 83.9 degrees). The landing point was reached at an angle of 83.9 degrees at c. 1327.0 m MD (1182.7 m TVD / -677.2 m SS), approximately equal to the original prognosed landing depth. The Sherwood porosity was encountered subsequent to re-entry after landing; at c. 1378.0 m MD (1187.0 m TVD / -681.5 m SS) 4.0 metres *below* the prognosed depth. The thickness of the interval between the Frobisher Evaporite top and the Sherwood porosity top at this location is approximately 8 metres greater than that of the prognosed thickness. Consequently the well was landed 4.3 metres above the Sherwood Porosity. Upon Landing the BHA was pulled out of the hole to change the PDC bit from a 222mm to a 200mm for the lateral section, and to dial back the motor to a 1.83 degree bend.

Subsequent to re-entry the BHA was rotated until a survey was acquired beyond the landing point. The BHA was then steered up (and for turn) to begin flattening off as the well bore approached the Sherwood porosity. The BHA was then rotated as it built angle. At approximately 1370 m MD the BHA suddenly began dropping angle. The BHA was steered up (and for turn) at c. 1374, 1384 and 1393 metres, with little result. Consequently the BHA continued to drop at c. 85 to 86 degrees as the top of the Sherwood porosity was transected. The BHA ascended approximately 2 metres below the top of the Sherwood porosity. (The upper most metre of the zone appeared to have the greatest potential as reservoir rock). The BHA was steered up more aggressively at c. 1403 and 1413 m MD. The more aggressive slides were effective and the BHA attained an angle of 90.4 degrees. The BHA was then rotated in anticipation of transecting the ‘better reservoir rock’ at the top of the zone. When the ‘better reservoir rock’ was not encountered by 1480 m MD, the decision was made to climb at 91 to 92 degrees to locate the zone. The ‘better reservoir’ zone was encountered at c. 1490 m MD, just as the BHA was steered up. At c. 1509, 1518 and 1526 the BHA was steered down to flatten off within the zone. The BHA was then rotated within the upper most

Sherwood porosity. At c. 1570 mMD the BHA was steered down to attain an angle below 90 degrees, but the well path continued to climb at a gentle angle. At c. 1590 mMD the BHA scraped the top of the Sherwood porosity and was subsequently steered down to an angle of 89.5 degrees. The remainder of the well was rotated within the upper-most Sherwood porosity. The leg was terminated at 1650.0 m MD (1188.0 m TVD / -682.4 mSS) considering the strong possibility losing tools down hole, due to common hole problems in this area.

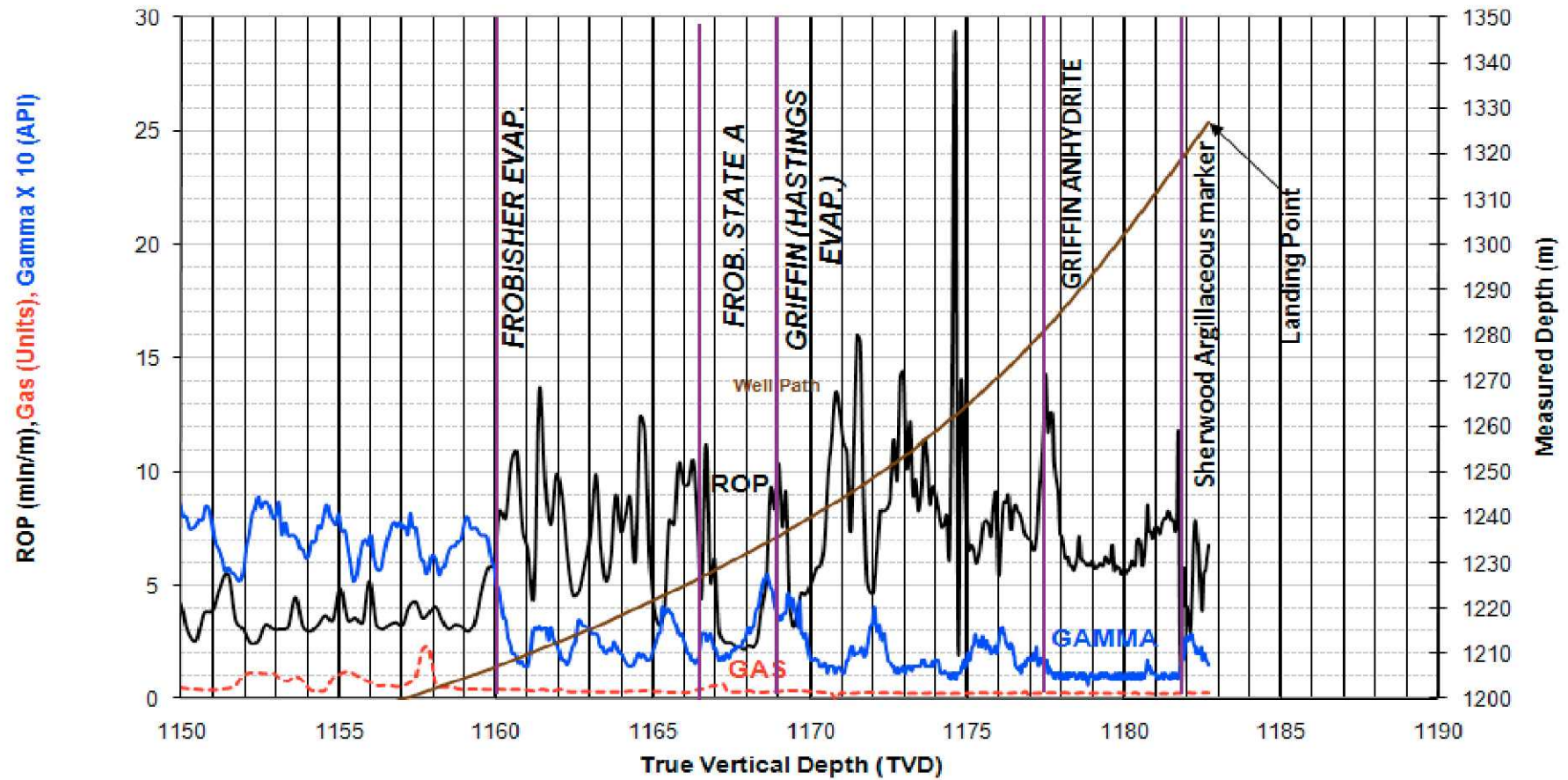
The target Sherwood porosity here comprises two main lithologies: The upper most Lithology is a cryptocrystalline to microcrystalline calcareous dolomite, finely interbedded with peloidal / skeletal wackestone to packstone, with fair intercrystalline (to lesser intergranular) porosity with patchy to uniform, light to medium brown oil stain, which exhibits a dull yellow fluorescence and slightly delayed streaming white cut. The second Lithology is a cryptocrystalline bioclastic to oolitic, and occasionally dolomitic wackestone, with generally poor intergranular (-intercrystalline matrix) porosity, and approximately 5-10% patchy light brown oil stain oil stain, which exhibits a light yellow fluorescence and a slow streaming white cut.

In summary, the well opened up a total of 325 metres of lateral section, comprising 105 metres of fair reservoir, 110 metres of poor to fair reservoir, 75 metres of poor reservoir, 15 metres of very poor reservoir, and 20 metres of non to very poor reservoir.

Jeff Bovee August 4, 2011

FCL et al Workman Hz 3B13-20/4B10-19-1-31W1M

BUILD SECTION: TVD vs MD



FCL et al Workman Hz 3B13-20/4B10-19-1-31W1M
ROP-Gas Summary

